

Enhancing Culinary Training with Spatial Augmented Reality: A User Study Comparing sAR Kitchen and Video Tutorials

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(a) A user experiencing sAR Kitchen

(b) sAR system in the user's view

Figure 1: sAR Kitchen

ABSTRACT

This paper presents sAR Kitchen, a cooking assistant that aims to enhance culinary training through the use of spatial augmented reality (sAR). We conducted a user study with twenty-two participants to investigate the effects of instructions given by our proposed sAR system and a monitor display with video tutorials in a playdough-making task representing the cooking task. Our study measured participants' perceived workload, usability, and performance regarding task completion time, product quality, and cooking station messiness. The results showed that sAR significantly decreased the perceived workload and increased the system's usability compared to traditional video-based tutorials. Additionally, our open-ended questions revealed that participants found the sAR instructions more engaging and interactive. Overall, this study demonstrates the potential of sAR to provide more user-friendly learning methods in culinary training.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools—User interface programming; Human-centered computing—Interaction design—Interaction design process and methods—Activity centered design

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1 INTRODUCTION

Augmented reality (AR) presents new ways of performing traditional tasks, especially in learning and training [5–7]. AR offers more freedom to create designs that better fit users' needs and can be used to present instructions for tasks, such as cooking [11]. In this study, we compared a video-based system with an AR-based system for delivering instructions for a cooking task, using spatial AR (sAR) technology. This system allows for the seamless integration of virtual elements on a real surface, and individuals can control the pace of the instruction for utmost learning [4]. This technology is versatile for work environments and beneficial for people who need more visual and interactive guidance.

AR technology has been used as an assistive tool for providing instructions in manufacturing tasks, such as assembly and disassembly [9, 13–15]. Although AR-based instructions improve task completion time, number of errors, and system usability, choosing the best technology for providing instructions remains unresolved. A study by [12] compared a projection-based AR system with AR smart glasses and found that the projection-based system reduced task completion time and error rate while also being more robust against lighting conditions. The comparison of the Spatial and Mobile AR for assembly procedure also showed that sAR is a more robust tool for improving task completion time, errors, and cognitive workload [1].

AR-based instructions have been used in culinary applications to enhance the user experience and make cooking less cognitively demanding. Previous studies have shown the effectiveness of AR for food safety training and cooking support [8, 10]. However, there is a need for a clear comparison of AR-based and video-based tutorials for simple tasks in everyday life. This current study aims to provide insight into optimizing technologies to support cognitive processes during sequential learning and to understand how multimedia cognitive learning theories can be integrated within workstations where manual pausing with a computer mouse or keyboard is not ideal.

Additionally, cognitive ergonomics and users' experience still need to be addressed. Thus, the hypotheses for this research are that using sAR for culinary training will increase usability (H1) and reduce perceived workload (H2) compared to a video-based approach. Additionally, it is hypothesized that sAR will improve performance, including task completion time, product quality, and cooking station messiness (H3).

2 METHODS

2.1 Participants

We recruited 22 participants including 12 males and 10 females, with varying experience levels in cooking and working with AR technology. The participants were between 18-64 years old, with an average age of 29.68. All participants had cooking experience. 8 of them reported that they cook regularly, and all of them had normal or corrected-to-normal vision. The Institutional Review Board reviewed and approved the research protocol (IRB# 2020-0466).

2.2 sAR System Design

A projection-based AR system was developed using a projector, and a desktop computer, a depth sensor camera, Microsoft Azure Kinect. The system uses augmented instructions projected onto a cooking station to guide users through a recipe. It allows for learner-paced segments, does not require tactile interactions, and was developed using Unity 2019. The system was designed to integrate digital content into a real-world kitchen setting seamlessly, and the Azure Kinect tracks the user's body and hands to enable interaction with the sAR user interface (UI). This aims to lower the barrier of entry to cooking and create a user-friendly experience. sAR Kitchen setup is visualized in Figure 1.

2.3 Experimental Design & Protocol

This study investigated the effect of the tutorial systems (AR-based vs. Video-based) on usability, workload, and performance (task completion time, product quality, and messiness of the cooking station). Using a within-subject design experiment, users performed a playdough-making task twice, once with each tutorial method, in a counterbalanced order to prevent bias. The procedure and goal of the study were explained to participants, who were asked to finish the task as quickly and accurately as possible. Before starting the experiment, participants were allocated 10 minutes to become familiar with the sAR interface. A timer was used to measure task completion time, and videos were recorded to measure the messiness of the cooking station. For each trial, participants were asked to make a playdough using the provided instructions. For the Video-based experiment, we showed them a pre-recorded video of the playdough-making task we designed, and for the AR-based task, the same video was played using sAR on the table. For the Video-based task, a desktop computer was placed in a normal position in front of the users, and participants were required to pause/replay/rewind the video to finish each step. However, the AR-based task was broken into steps, and participants were required to hover over the Next button once they were done with each step. They could also hover over the Back button if they needed to replay a step. Moreover, since participants were seated while doing the tasks, they were asked to adjust the chair height in each session to be in the best position to watch the videos on both systems. Participants completed questionnaires (NASA Task Load Index [3], and System Usability Scale [2]) after each trial. The experiment took about 60 minutes, and participants were compensated for their time. After the experiment, they were asked to provide their thoughts and feedback on the benefits and drawbacks of the two systems through open-ended questions. The final playdough product and the messiness of the cooking station were rated by two independent judges following the completion of the experiment.

3 RESULTS

3.1 NASA-TLX

The results of the Kruskal-Wallis test indicate that there is a significant difference between the two systems in terms of Mental Demand ($X^2 = 3.827$, $p = 0.049$), Temporal Demand ($X^2 = 20.204$, $p = 0.0006$), Effort ($X^2 = 6.64$, $p = 0.009$), Performance ($X^2 = 15.783$, $p = 0.0007$), and Frustration ($X^2 = 4.342$, $p = 0.03$). The participants gave lower scores to the sAR Kitchen in these questions, as it was found to be less mentally and temporally demanding, less frustrating, required less effort, and led to better task performance. However, there was no significant difference between the systems regarding Physical Demand, as both required the users to be seated at the cooking station, and there were no major differences in physical movements. The NASA-TLX plot is shown in Figure 2.

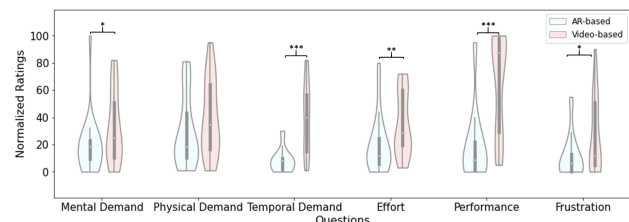


Figure 2: NASA-TLX score of the two systems

3.2 System Usability Scale (SUS)

The results of the t-test on the SUS scores of each system indicate that there is a significant difference between the systems ($t(21) = 2.080$, $p = 0.034$). The sAR Kitchen outperformed the Video-based system in usability. The mean score of the total SUS for the sAR and the Video-based system is 83.30 and 75.34, respectively. Generally, a score of 68 or higher is considered good for SUS. Although both of the systems were rated as usable, a significantly higher score for sAR shows that compared to the conventional Video-based method, this system is better in different dimensions of effectiveness, efficiency, and overall ease of use. The SUS plot is shown in Figure 3.

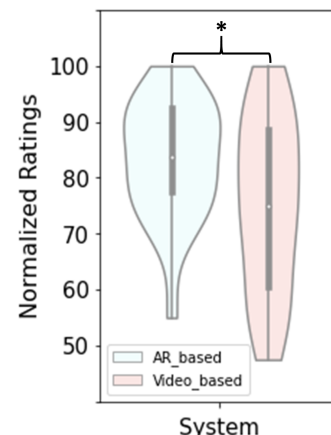


Figure 3: Total SUS score of the two systems

3.3 Performance Measures

Task completion times for the AR-based system ranged from 333 to 974 seconds ($M = 616.81$, $SD = 221.84$), and for the Video-based system, from 418 to 1,152 seconds ($M = 658$, $SD = 200.445$). Although the Kruskal-Wallis test showed no significant difference in

task completion times for both systems ($X^2 = 0.753$, $p = 0.385$), sAR kitchen performed better according to descriptive data. The statistical analysis results, using the chi-squared test, did not reveal any significant difference in the quality of the final products between the two systems ($X^2 = 0.102$, $p = 0.747$). Additionally, the analysis also revealed no significant difference in the messiness of the cooking station between the two systems ($X^2 = 0.415$, $p = 0.336$).

3.4 Qualitative Feedback

Regarding the sAR Kitchen, 18 participants had a positive experience and highlighted the ease of use and smoothness. They noted that the sAR was interactive and engaging and allowed them to complete the task at their own pace without needing to touch the UI. They found that the pace of the task in the Video-based system was faster and the sAR allowed them to have full control over the UI, which made the task easier and more comfortable. They also stated that the sAR Kitchen reduced the need for head movement compared to the Video-based system, and that they preferred the sAR Kitchen over the Video-based system if they had the chance to redo the task. However, they noted that the sAR is not widely available yet.

4 DISCUSSION

In this study, the sAR instructions showed better results than the Video-based instructions. The qualitative feedback from participants supported the statistical findings. For usability, the sAR Kitchen had a significant advantage over the Video-based system, as it was found to be easier and more well-integrated to use. However, it was noted that a technical person was needed to teach them how to use the system. They also found the sAR Kitchen less demanding and more comfortable to use. The results supported the first and second hypotheses, but the third hypothesis was not fully supported due to non-significant results for product quality and cooking station messiness. The participants expressed positive feedback towards using sAR and willingness to use it frequently. Using the sAR system benefited the user's engagement without needing other obtrusive devices or disturbing cooking stations. In general, sAR system outperformed Video-based tutorials and has the potential for real-world applications.

5 CONCLUSION, LIMITATIONS AND FUTURE DIRECTIONS

This study compared a spatial AR system and a Video-based method for providing instructions for a playdough-making task, representing a cooking task in culinary training applications. The results showed that users preferred the sAR Kitchen for usability and workload. The study highlighted the benefits of AR technology for traditional tasks such as cooking and the shortcomings of the Video-based method. However, it should be noted that the limitations of lab-based experiments and using a similar task and environment to the cooking condition may have affected the results. For this experiment, due to the location of the Azure sensor for a restricted lab environment, we asked users to be seated while working on the task to allow hand tracking to track their hands accurately. Although none of the users mentioned that this was limiting their movements, it should be considered a limitation. Future research will focus on extending the AR system to detect kitchenware, providing relevant information and instruction, and improving usability through adaptable UI. This technology can improve user experience, prevent potential injuries, especially for training purposes with novice users, and improve users' overall well-being and safety. This type of interactive sAR system allows people to learn without being required to touch/test before learning. Additional research methods such as cognitive or physiological measurements, eye-tracking technology, ethnography, and field studies can further evaluate AR technology's use in real-life scenarios and assist people with disabilities. A technology acceptance study can also be conducted to validate the acceptability of such technology in working environments.

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